EFY TESTED
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'NAMASTE' ROBOT
To Greets Visitors Automatically

NAVPREET SINGH TUNG

This robot, whose mechanical frame can be made of recycled materials, monitors people within its 100cm range using an ultrasonic module. Detecting anyone within this range, it greets the person with 'Namaste' symbolised by both hands joined together in traditional Indian greeting. The robot is versatile and can be deployed in diverse settings such as business locations, parks, shops, etc, to extend greetings. Despite its primary function, the robot is designed to inspire modification for additional purposes. For instance, envision the potential for adapting its technology for use near automatic gates at airports or garages.

Users have the flexibility to substitute the servo with various actuators for functions like automatically opening gates or doors. Alternatively, a larger servo may be attached to actuate gates or doors for automatic opening. The device is adaptable and can be repurposed for various applications. However, our focus remains on demonstrating the sensing of human presence, activating the servo motor to move the robot's hands and welcome people when someone approaches. For robot's mechanical hand, users can purchase or 3D print a robot hand like the inMoov robot hand, or design a custom robot structure and body. Fig. 1 shows the author's prototype of the inMoov hand.

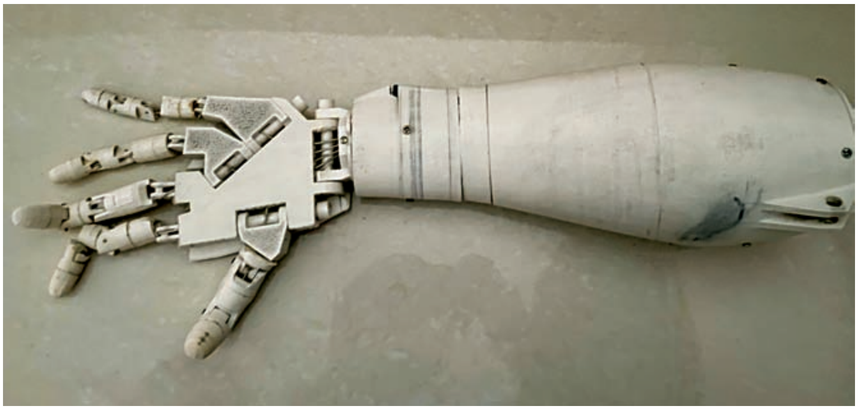

Fig. 1: InMoov robot hand

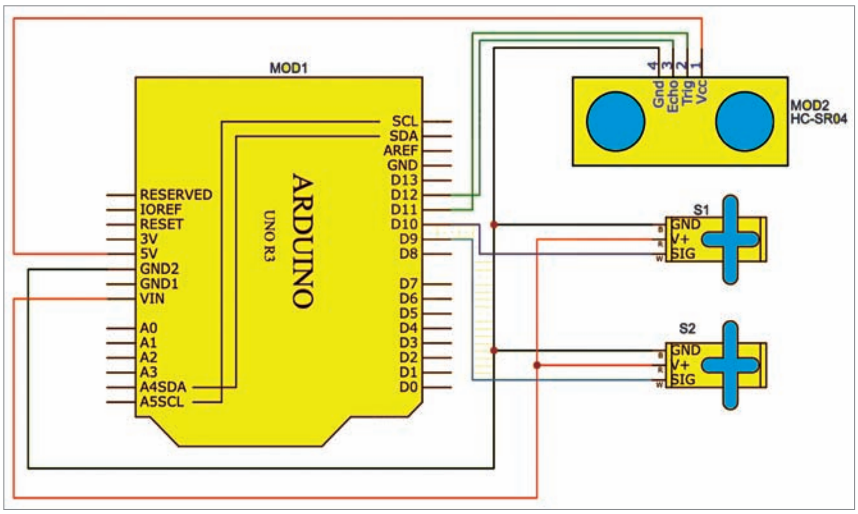

Fig. 2: Circuit diagram

Circuit and working

Fig. 2 illustrates the circuit diagram of the device, which is based on Arduino Uno technology. It comprises Arduino Uno (MOD1), an ultrasonic sensor HC-SR04 (MOD2), two SG90 servo motors (S1, S2), a mechanical body from recycled materials, a breadboard, and jumper wires. Users have the option to replace the Arduino Uno with other boards such as Arduino Mega or Arduino Nano. Brief introduction to the compo-

BILL OF MATERIALS

Components	Quantity
Arduino Uno/Mega/Nano (MOD1)	1
Servo motor SG90 (S1, S2)	2
Ultrasonic sensor HC-SR04 (MOD2)	1
Mechanical body for robot	Optional
Jumper wires (male-to-male)	12
Breadboard	1